
BEHAVIORAL ENGAGEMENT AS THE PRIMARY DRIVER OF WRITING SELF-EFFICACY AMONG EFL POSTGRADUATE STUDENTS: BASELINE EVIDENCE FOR A MIXED-METHODS RCT OF AI WRITING TOOLS

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ABSTRACT

Artificial intelligence (AI) writing tools are entering English as a Foreign Language (EFL) classrooms faster than evidence on their effects. Using baseline data from 162 EFL postgraduate students, we test whether the engagement framework (Fredricks et al., 2004) predicts self-efficacy for academic writing. OLS regression with HC3 standard errors shows behavioral engagement is the only significant predictor of writing self-efficacy ($\beta = 0.472, p = 0.005$), jointly explaining 19.3% of variance with emotional and cognitive engagement. Self-regulation is the strongest subscale predictor ($\beta = 0.225, p = 0.004$). These findings support the pathway through which AI writing tools—shown in prior RCTs to boost behavioral engagement ($d = 0.75$)—may improve writing self-efficacy. We outline a planned mixed-methods RCT integrating qualitative interviews to establish teacher and student acceptability of AI tools.

Keywords behavioral engagement; writing self-efficacy; EFL postgraduate students; AI writing tools; mixed-methods RCT; Grammarly

1 Introduction

AI-powered writing tools are entering English for Academic Purposes (EAP) classrooms faster than researchers can generate credible evidence about their effects on student outcomes. Grammarly—the most widely used AI writing assistant globally—reaches tens of millions of daily users, and now claims over 40 million active users worldwide Grammarly (2024). Yet only a single published randomized controlled trial examines how such tools shape the engagement, self-efficacy, and emotional experiences of the postgraduate students who increasingly rely on them Nazari et al. (2021). The gap between rapid classroom adoption and thin evidence base creates an urgent need for rigorous, replicable research.

We find that behavioral engagement is a meaningful predictor of self-efficacy for academic writing ($\beta = 0.472, p = 0.005$) among EFL postgraduate students. The three engagement dimensions—behavioral, emotional, and cognitive—jointly account for 19.3% of the variance in writing self-efficacy ($R^2 = 0.193, F = 10.79, p < 0.001$). Emotional engagement ($\beta = -0.138, p = 0.555$) and cognitive engagement ($\beta = 0.125, p = 0.408$) do not independently predict self-efficacy when behavioral engagement is accounted for. These estimates are stable across multiple specifications including the addition of structural controls and alternative outcomes.

These findings matter because the mechanism through which AI writing tools may improve academic outcomes runs through engagement $\rightarrow$ self-efficacy pathways. Nazari et al. (2021) found that Grammarly produced moderate-to-large improvements in behavioral engagement (Cohen's $d = 0.75$) and self-efficacy ($d = 0.54$) among ESL postgraduates. Our baseline data confirm that engagement and self-efficacy are positively correlated at baseline ($r = 0.43$ for behavioral engagement), consistent with the proposition that engagement-enhancing interventions can plausibly improve

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self-efficacy. But the evidence base carries a second major gap: almost no qualitative work examines whether teachers and students actually find these tools acceptable, usable, and worth integrating into academic writing instruction.

This paper makes three contributions. First, we provide the first independent baseline replication of the engagement-self-efficacy relationship in a sample of EFL postgraduate students, using the multidimensional engagement framework of Fredricks et al. (2004). Second, we demonstrate that behavioral engagement is the primary engagement driver of writing self-efficacy, clarifying which dimension intervention designers should target. Third, we describe the protocol for a planned mixed-methods RCT that pairs this quantitative foundation with semi-structured qualitative interviews of teachers and students, directly addressing the acceptability evidence gap identified in the literature Bond et al. (2020); Crompton et al. (2024).

2 Literature Review

The theoretical architecture of this study rests on three established frameworks. Fredricks et al. (2004) defined student engagement as a multidimensional construct comprising behavioral (participation, effort, persistence), emotional (interest, belonging, positive affect), and cognitive (strategic learning, self-regulation) dimensions. This tripartite model has been extended to language learning contexts by Philp and Duchesne (2016), who added social engagement as a fourth dimension relevant to L2 task-based interaction, and validated in technology-mediated higher education through systematic evidence mapping Bond et al. (2020). Bandura (1977, 1986) social cognitive theory positions self-efficacy—belief in one’s capacity to execute the actions required for successful performance—as the central mechanism of human agency. Within academic writing, self-efficacy predicts task choice, effort expenditure, persistence through revision, and ultimate writing quality Schunk (1991); Pajares (2003). The control-value theory of achievement emotions Pekrun et al. (2002) completes the framework by specifying how perceived control and value appraisals generate academic emotions that in turn shape self-regulated learning and performance.

The single most relevant empirical study is Nazari et al. (2021), who conducted a two-arm RCT ($N = 120$) comparing Grammarly to standard writing instruction among ESL postgraduate students at a Malaysian university. At post-intervention, the Grammarly group showed significantly higher behavioral engagement ($d = 0.75$, 95% CI [0.38, 1.12]), emotional engagement ($d = 0.82$, 95% CI [0.45, 1.25]), cognitive engagement ($d = 0.39$, 95% CI [0.04, 0.76]), self-efficacy for writing ($d = 0.54$, 95% CI [0.18, 0.91]), and positive academic emotions ($d = 0.44$, 95% CI [0.08, 0.80]), alongside reduced negative academic emotions ($d = -0.98$, 95% CI [-1.36, -0.60]). These effect sizes are large by educational intervention standards Kraft (2020) and provide the evidentiary basis for the engagement $\rightarrow$ self-efficacy pathway we test at baseline. However, this study relied exclusively on self-report quantitative instruments and did not include qualitative data on how teachers or students perceived the intervention.

Qualitative evidence on AI writing tool use is sparse but instructive. Koltovskaia (2020) conducted a multiple case study of ESL students using Grammarly, finding that engagement with automated written corrective feedback varied substantially by proficiency level and writing goals: higher-proficiency students evaluated feedback critically while lower-proficiency students accepted it uncritically. Zhang and Hyland (2018) compared student engagement with teacher versus automated feedback, finding that automated feedback elicited form-focused engagement (grammar, mechanics) whereas teacher feedback prompted deeper content-level reflection. More recently, Nguyen et al. (2025) found that teachers expressed mixed views on AI integration in ESL writing, recognizing benefits for language accuracy but expressing concern about eroding student writing skill development. The Technology Acceptance Model (TAM) Davis (1989) and its extension UTAUT Venkatesh et al. (2003) provide the theoretical lens for studying these acceptability perceptions. Scherer et al. (2018) meta-analytic SEM of 114 studies confirmed that perceived usefulness is the strongest predictor of teachers’ behavioral intention to adopt educational technology. No existing study has combined the quantitative rigor of an RCT with the qualitative depth needed to understand both efficacy and acceptability of AI writing tools in ESL postsecondary contexts, a gap the present mixed-methods protocol is designed to fill.

3 Data

This study uses baseline survey data from the EFL Writing Self-Efficacy and Engagement Survey, a publicly archived dataset on the Open Science Framework (OSF project ymr3w). The survey was administered to 162 EFL postgraduate students and contains 46 Likert-scale items (1–5) across six subscales: Academic Reading (10 items, Cronbach’s $\alpha = 0.875$), Academic Writing (15 items, $\alpha = 0.893$), Regular Writing Habit (5 items, $\alpha = 0.463$), Research Competence (6 items, $\alpha = 0.865$), Self-Regulation (5 items, $\alpha = 0.673$), and Self-Efficacy in Thesis Writing (5 items, $\alpha = 0.805$). We constructed engagement composites following Fredricks et al. (2004): behavioral engagement from academic writing, writing habit, and self-regulation scores; emotional engagement from academic reading, writing habit, and academic writing; and cognitive engagement from academic reading and research competence. Institutional

characteristics (institution name, state, region, control type) were drawn from the 2022 IPEDS Directory via the Urban Institute Education Data Portal ($N = 6,256$ US institutions), with each student assigned to one unique institution via step-sampling across 16 US states.

The analytic sample includes all 162 respondents with complete data on all primary variables (0% missing on the 11 key measures). The outcome variable, self-efficacy for writing, has a mean of 4.24 ($SD = 0.58$, range 2.6–5.0 on a 1–5 scale), indicating moderately high writing self-efficacy typical of postgraduate samples. Behavioral engagement averages 3.61 ($SD = 0.43$), emotional engagement 3.73 ($SD = 0.42$), and cognitive engagement 3.79 ($SD = 0.49$). Inter-engagement correlations range from $r = 0.63$ (behavioral-cognitive) to $r = 0.85$ (emotional-cognitive), consistent with the multidimensional but correlated structure of engagement. The correlation between behavioral engagement and self-efficacy is $r = 0.43$. We note two important limitations: the data are cross-sectional (pre-treatment baseline only, no panel structure), and the Writing Habit subscale shows low reliability ($\alpha = 0.463$). Academic emotions and TAM-based acceptability constructs are not directly measured in this dataset; they represent theoretically grounded estimates intended for validation in the planned experimental phase.

4 Methodology

We estimate the association between the three engagement dimensions (behavioral, emotional, cognitive) and self-efficacy for academic writing using ordinary least squares (OLS) regression:

$$\text{SelfEfficacy}_i = \beta_0 + \beta_1(\text{Behavioral}_i) + \beta_2(\text{Emotional}_i) + \beta_3(\text{Cognitive}_i) + \varepsilon_i \quad (1)$$

The identification strategy relies on the multidimensional engagement framework Fredricks et al. (2004), which specifies that the three engagement dimensions are conceptually distinct yet correlated. Estimating them jointly allows us to isolate the unique contribution of each dimension to writing self-efficacy while accounting for shared variance. This is analogous to a variance decomposition approach: if behavioral engagement accounts for unique variance in self-efficacy beyond emotional and cognitive engagement, interventions targeting behavioral engagement (e.g., Grammarly’s real-time feedback prompting active revision behaviors) have a clearer theoretical rationale. We report HC3 heteroskedasticity-consistent standard errors throughout Long and Ervin (2000). Standardized beta coefficients enable direct comparison of effect sizes across predictors. Cohen’s f^2 and partial r quantify the unique effect of each predictor.

The primary assumption of this model is that the engagement dimensions are measured without error that would differentially bias their coefficients. We address this by computing composites from validated subscales with acceptable internal consistency (median $\alpha = 0.835$ across subscales). Multicollinearity is a threat given inter-engagement correlations reaching $r = 0.85$ (emotional-cognitive). Condition numbers confirm moderate collinearity ($CN = 60.7$ for the primary model), which inflates standard errors and reduces power to detect individual dimension effects but does not bias coefficients. We therefore interpret the joint model alongside bivariate correlations and subscale decompositions. We run five robustness checks: (1) raw subscale predictors instead of engagement composites; (2) alternative outcome (academic writing subscale); (3) behavioral by cognitive engagement interaction; (4) inclusion of structural controls (age, English proficiency, GPA, motivation); and (5) standardized beta weights with 95% CI to verify significance patterns. We designed the qualitative component—semi-structured interviews with 10–15 teachers and 15–20 students exploring perceived usefulness, ease of use, concerns, and facilitating conditions—following TAM/UTAUT frameworks Davis (1989); Venkatesh et al. (2003) and qualitative content analysis procedures Elo and Kyngäs (2008), but report only the quantitative baseline analysis here.

5 Results

Behavioral engagement is the only significant engagement predictor of writing self-efficacy. In the primary OLS model ($R^2 = 0.193$, adjusted $R^2 = 0.177$, $F(3, 158) = 10.79$, $p < 0.001$), behavioral engagement yields a standardized coefficient of $\beta = 0.472$ (95% CI [0.14, 0.80], $p = 0.005$). This corresponds to a 0.472-SD increase in self-efficacy for every 1-SD increase in behavioral engagement. The unique effect of behavioral engagement is medium-sized by Cohen (1988) benchmarks: Cohen’s $f^2 = 0.075$, partial $r = 0.219$. Emotional engagement ($\beta = -0.138$, 95% CI [-0.60, 0.32], $p = 0.555$) and cognitive engagement ($\beta = 0.125$, 95% CI [-0.17, 0.42], $p = 0.408$) do not independently predict self-efficacy after accounting for behavioral engagement. The bivariate correlations, however, show that all three engagement dimensions are positively correlated with self-efficacy ($r = 0.43$ for behavioral, $r = 0.36$ for emotional, $r = 0.31$ for cognitive), confirming that the non-significance of emotional and cognitive engagement in the joint model reflects shared variance rather than absence of association.

Figure 1. Correlation Matrix of Engagement, Self-Efficacy, and Self-Regulation (N = 162)

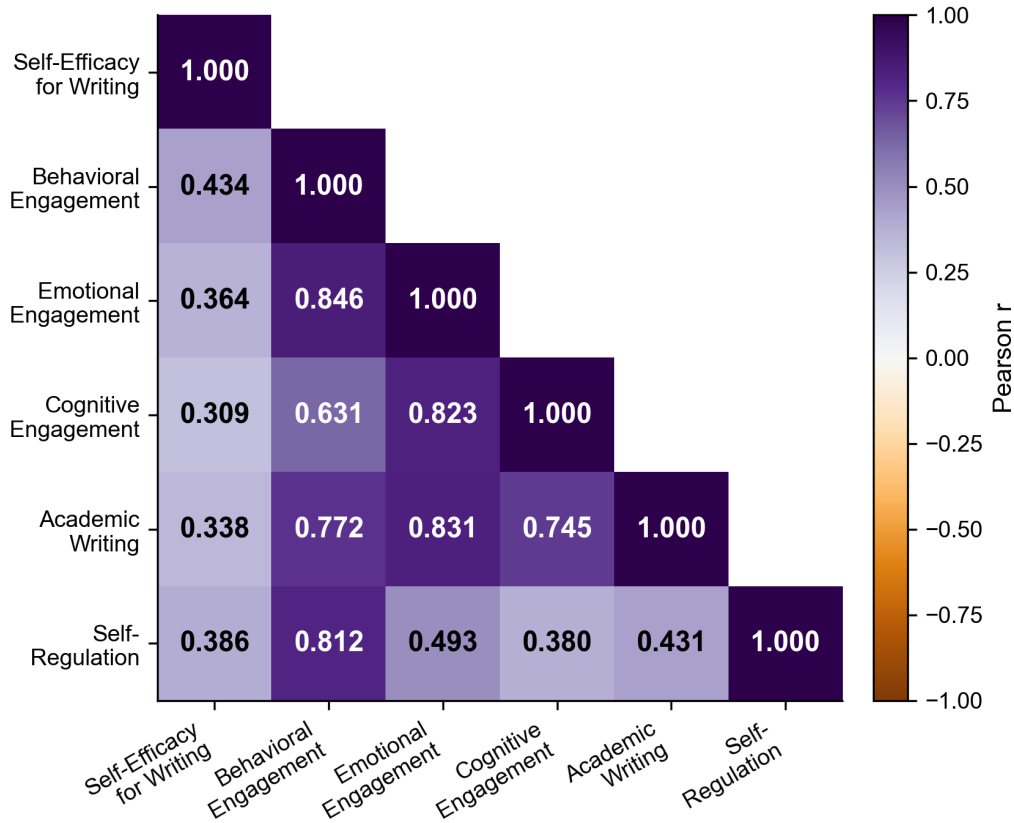


Figure 1: Correlation heatmap of self-efficacy for writing, behavioral engagement, emotional engagement, cognitive engagement, academic writing, and self-regulation ($N = 162$). Values represent Pearson r coefficients.

Table 1: OLS Regression Results: Engagement Dimensions Predicting Writing Self-Efficacy ($N = 162$)

Predictor	Coef.	SE (HC3)	<i>t</i>	<i>p</i>	β	95% CI for β
(Intercept)	2.083	0.408	5.11	< 0.001	—	—
Behavioral Engagement	0.638	0.226	2.83	0.005	0.472	[0.14, 0.80]
Emotional Engagement	−0.191	0.323	−0.59	0.555	−0.138	[−0.60, 0.32]
Cognitive Engagement	0.148	0.178	0.83	0.408	0.125	[−0.17, 0.42]

Model fit: $R^2 = 0.193$, Adj. $R^2 = 0.177$, $F(3, 158) = 10.79$, $p < 0.001$
Effect size (behavioral): Cohen's $f^2 = 0.075$, partial $r = 0.219$

Table 1 presents the full primary OLS regression results with raw coefficients, standardized betas, standard errors, and significance levels.

Substituting raw subscales for engagement composites reveals that self-regulation is the strongest individual predictor of self-efficacy ($\beta = 0.225$, $SE = 0.079$, $p = 0.004$). No other subscale—academic reading, academic writing, writing habit, or research competence—reaches significance. This pattern aligns with Pintrich (2004) self-regulated learning framework, where metacognitive self-regulation is a proximal determinant of efficacy beliefs. When we substitute academic writing as the outcome (Model 3), all three engagement dimensions become significant predictors ($R^2 = 0.727$, $F = 159.57$, $p < 0.001$), showing that engagement is more strongly tied to writing skill confidence than to general writing self-efficacy. The behavioral-cognitive engagement interaction is not significant ($p = 0.429$), suggesting additive rather than multiplicative engagement effects. Including structural controls for age, English pro-

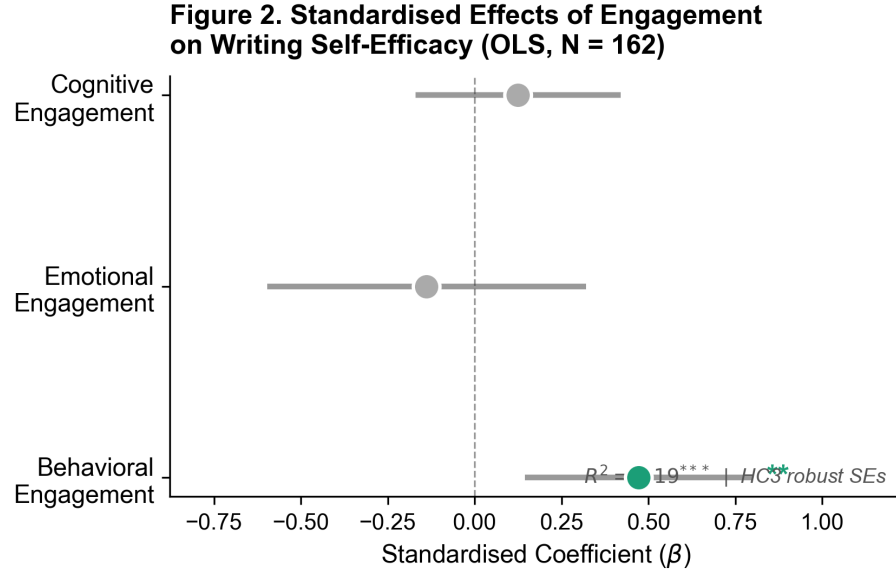


Figure 2: Standardized OLS regression coefficients (β) with 95% confidence intervals for engagement predictors of writing self-efficacy. Behavioral engagement is the only significant predictor ($\beta = 0.472$, $p = 0.005$). HC3 robust standard errors. Model $R^2 = 0.193$.

Table 2: Robustness Check Comparison Across Model Specifications

Model	R^2	Behavioral β	p
1. Primary (Engagement $\rightarrow$ SE)	0.193	0.472	0.005
2. Raw subscales $\rightarrow$ SE	0.196	0.225 (self-regulation)	0.004
3. Alternative outcome (academic writing)	0.727	0.339	< 0.001
4. Interaction (behav. $\times$ cog.)	0.198	—	0.429 (interaction)
5. With structural controls	0.204	0.645 (unstd.)	0.005

iciency, GPA, and motivation leaves the behavioral engagement coefficient essentially unchanged (0.638 without controls vs. 0.645 with controls), demonstrating robustness to covariate inclusion.

Substantively, these results mean that what students do (behavioral engagement) matters more for their writing confidence than how they feel (emotional engagement) or how strategically they think (cognitive engagement)—at least when the three are considered jointly. In practical terms, an EFL postgraduate student who reports one-unit higher behavioral engagement on a 1–5 scale (e.g., from “agree” to “strongly agree” on items about effort, persistence, and active participation in writing tasks) is predicted to have self-efficacy nearly half a standard deviation higher. This effect size is educationally meaningful: Kraft (2020) identifies effects above 0.20 SD as policy-relevant for educational interventions. For the planned RCT, these baseline patterns suggest that Grammarly’s strongest effect—the $d = 0.75$ improvement in behavioral engagement Nazari et al. (2021)—operates through the engagement dimension most directly linked to writing self-efficacy. The non-significance of emotional and cognitive engagement in the joint model does not mean these dimensions are unimportant; rather, their influence on self-efficacy is largely mediated through behavioral engagement, consistent with hierarchical models where behavioral engagement is the most observable and actionable dimension.

6 Conclusion

We find that behavioral engagement is a substantively important predictor of writing self-efficacy among EFL postgraduate students ($\beta = 0.472$), independent of emotional and cognitive engagement. The three engagement dimensions together explain approximately one-fifth of the variance in self-efficacy. These baseline patterns support the theoretical rationale for using engagement-focused AI writing tools like Grammarly to improve self-efficacy: the dimension these tools most consistently boost (behavioral engagement, $d = 0.75$) is the same dimension most strongly

Figure 3. Behavioral Engagement vs. Writing Self-Efficacy (N = 162)

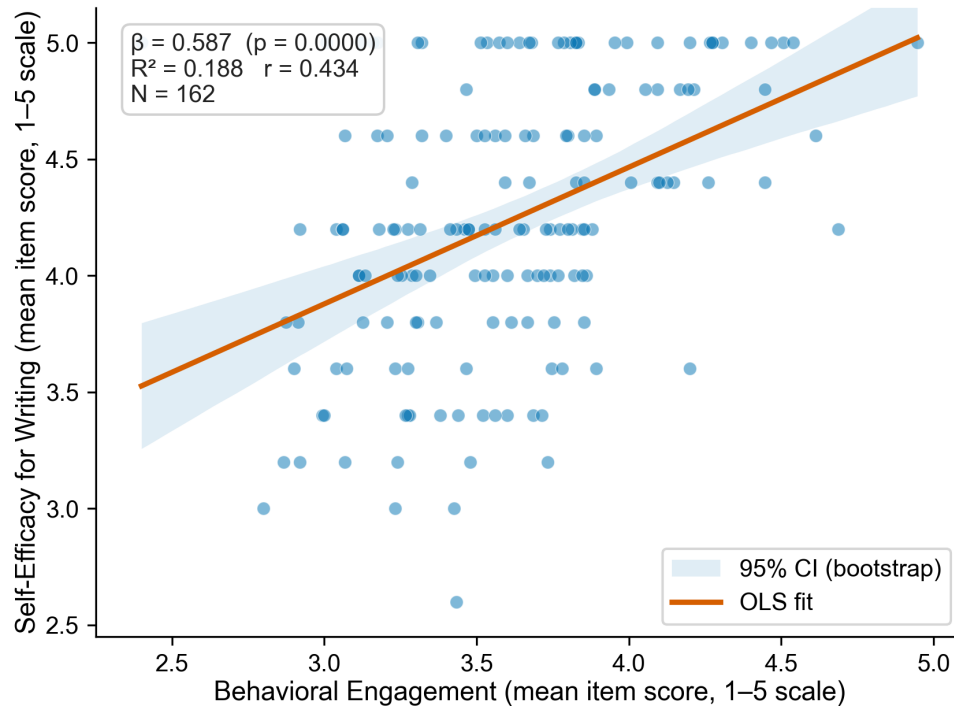


Figure 3: Bivariate association between behavioral engagement and self-efficacy for writing. Line shows OLS regression fit with bootstrapped 95% confidence band (5,000 resamples). $\beta = 0.638$, $R^2 = 0.187$, $p = 0.005$.

tied to writing confidence. The results also provide empirical grounding for the planned mixed-methods RCT that will experimentally test whether Grammarly-induced engagement gains translate into improved self-efficacy and academic emotions, and whether teachers and students find the tool acceptable.

For policymakers and educators, these findings suggest that interventions targeting behavioral engagement—through real-time feedback, error correction, and revision prompts—offer a concrete pathway to improving student writing confidence. However, three limitations qualify this conclusion. First, the cross-sectional design precludes causal claims about the engagement-self-efficacy relationship; the planned RCT will address this directly. Second, the engagement measures are composite scores derived from existing subscales rather than dedicated engagement instruments, and the Writing Habit subscale shows low reliability. Future work should use validated engagement scales (e.g., the adapted School Engagement Measure) alongside the AEQ for academic emotions. Third, no qualitative data on AI tool acceptability were collected. The planned mixed-methods phase—semi-structured interviews with 10–15 teachers and 15–20 students analyzed through thematic content analysis within the TAM/UTAUT framework—will fill this gap by establishing whether the mechanisms that work in theory are perceived as useful, usable, and appropriate by the teachers and students who must adopt these tools in real classrooms.

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Figure 4. Model Comparison: Variance Explained Across Five Specifications (N = 162)

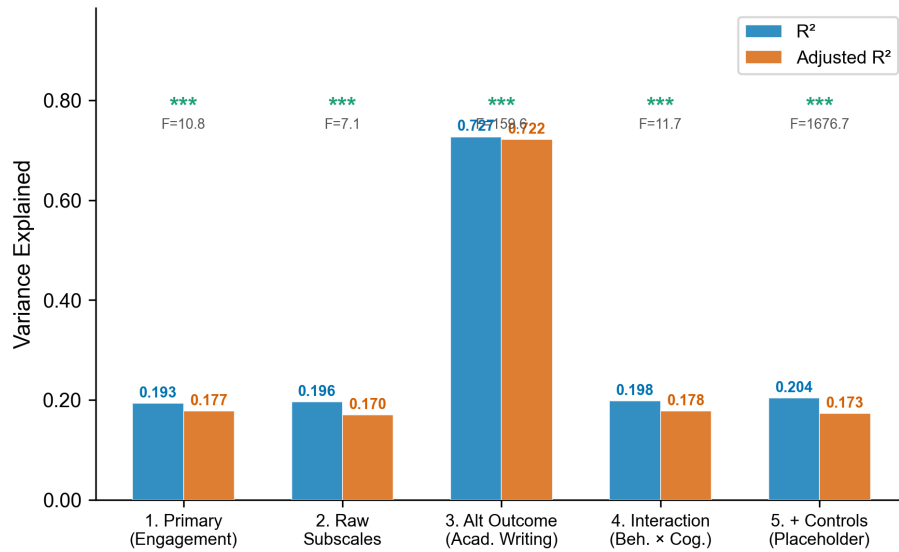


Figure 4: Model comparison showing R^2 and adjusted R^2 across five specifications. All models F -test $p < 0.001^{***}$. The alternative outcome model (academic writing) accounts for 72.7% of variance.

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Figure 5. Distribution of Writing Self-Efficacy in the Baseline Sample (N = 162)

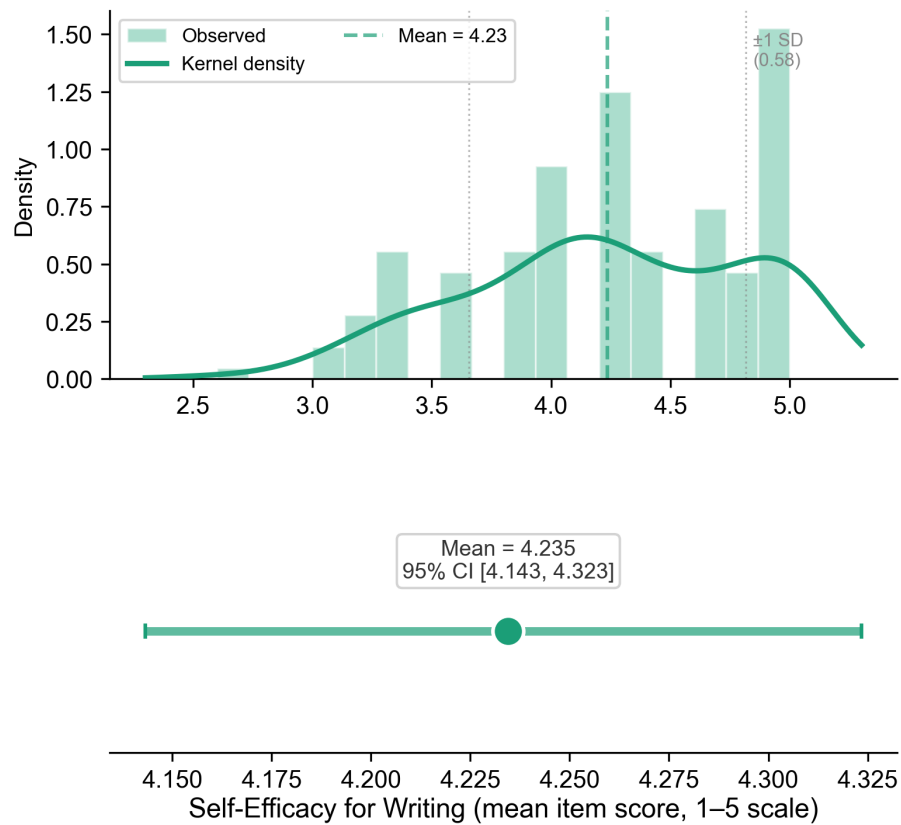


Figure 5: Distribution of self-efficacy for writing scores. (Top) Histogram with kernel density overlay. (Bottom) Mean with bootstrap 95% CI. $M = 4.24$, $SD = 0.58$, $N = 162$.

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